Example: Attribute Selection with Information Gain

- Class P: buys_computer = "yes"
- Class N: buys_computer = "no"

$$Info(D) = I(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940$$

age	p_i	n_i	$I(p_i, n_i)$
<=30	2	3	0.971
31…40	4	0	0
>40	3	2	0.971

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

$$Info_{age}(D) = \frac{5}{14}I(2,3) + \frac{4}{14}I(4,0) + \frac{5}{14}I(3,2) = 0.694$$

$\frac{5}{14}I(2,3)$ means "age <=30" has 5 out of 14 samples, with 2 yes'es and 3 no's.

Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

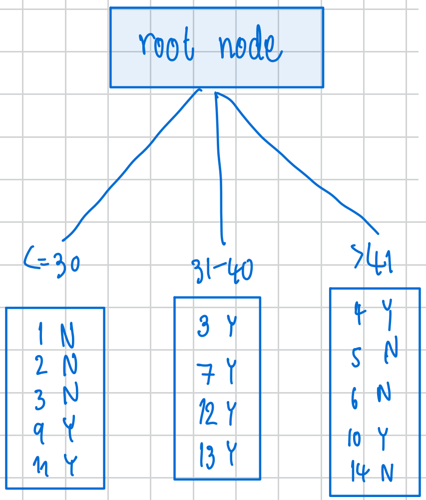
Similarly, we can get

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit\_rating) = 0.048$$

$$Info(D) = I(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940$$



age	p_i	n_i	$I(p_i, n_i)$
<=30	2	3	0.971
31…40	4	0	0
>40	3	2	0.971

$$I(2,3) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right) = 0.971$$

$$I(4,0) = -\frac{4}{4}\log_2\left(\frac{4}{4}\right) - \frac{0}{4}\log_2\left(\frac{0}{4}\right) = 0$$

$$I(2,3) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right) = 0.971$$

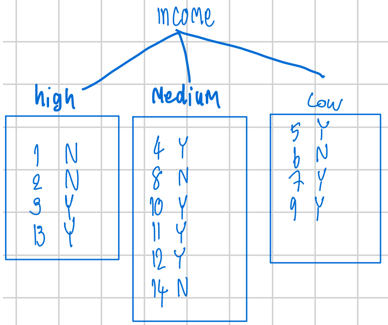
$$Info_{income}(D) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$$

$$Gain_{Age} = 0.940 - 0.694$$
$$= 0.246$$

$$Info(D) = I(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940$$

Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31…40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31…40	medium	no	excellent	yes
13	31…40	high	yes	fair	yes
14	>40	medium	no	excellent	no



Income	p_i	n_i	$I(p_i, n_i)$
high	2	2	1
medium	4	2	0.9182
low	3	1	0.8113

$$I(2,2) = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right) = 1$$

$$I(4,2) = -\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2\left(\frac{2}{6}\right) = 0.9182$$

$$I(3,1) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) = 0.8113$$

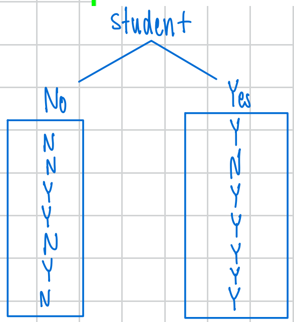
$$Info_{income}(D) = \frac{4}{14}(1) + \frac{6}{14}(0.9182) + \frac{4}{14}(0.8113) = 0.9110$$

$$Gain(income) = 0.940 - 0.9110$$

$$= 0.029$$

Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no



Student	p_i	n_i	$I(p_i, n_i)$
No	3	4	0.9852
Yes	6	1	0.5917

$$I(3,4) = -\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{4}{7}\log_2\left(\frac{4}{7}\right)$$

$$I(6,1) = -\frac{6}{7}\log_2\left(\frac{6}{7}\right) - \frac{1}{7}\log_2\left(\frac{1}{7}\right)$$

$$Info_{student}(D) = \frac{7}{14}(0.9852) + \frac{7}{14}(0.5917)$$

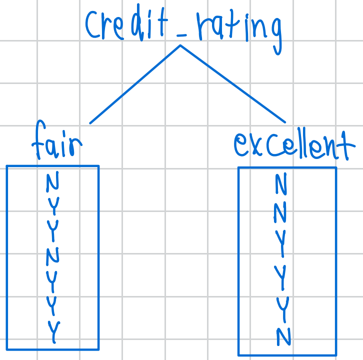
$$= 0.7885$$

$$Gain_{student} = 0.940 - 0.7885$$

$$= 0.1515$$

Training data set: Who buys computer?

age	income	studen	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no



Credit	p_i	n_i	$I(p_i, n_i)$
fair	6	2	0.8113
excellent	3	3	1

$$I(6,2) = -\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right)$$

$$I(3,5) = -\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{5}{6}\log_2\left(\frac{5}{6}\right)$$

$$Info_{credit} = \frac{8}{14}I(6,2) + \frac{6}{14}I(3,3)$$

$$= \frac{8}{14}(0.8113) + \frac{6}{14}(1)$$

$$= 0.8922$$

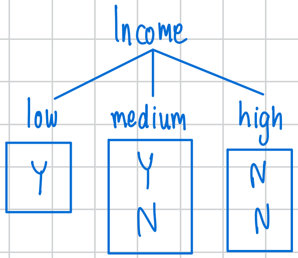
$$Gain_{credit} = 0.940 - 0.8922$$

$$= 0.0478$$

* root แรก

$$Info(D) = I(2,3) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right) = 0.9701$$

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes



Income	p_i	n_i	$I(p_i,n_i)$
high	1	0	0
medium	1	1	1
low	0	2	0

$$I(1,0) = -\frac{1}{1}\log_2\left(\frac{1}{1}\right) - \frac{0}{1}\log_2\left(\frac{0}{1}\right) = 0$$

$$I(1,1) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$$

$$I(0,2) = -\frac{0}{2}\log_2\left(\frac{0}{2}\right) - \frac{2}{2}\log_2\left(\frac{2}{2}\right) = 0$$

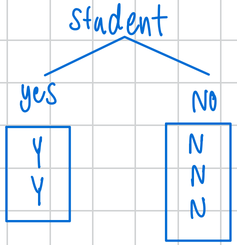
$$Info_{income}(D) = \frac{1}{5}I(1,0) + \frac{2}{5}I(1,1) + \frac{2}{5}I(0,2)$$

$$= \frac{1}{5}(0) + \frac{2}{5}(1) + \frac{2}{5}(0)$$

$$= 0.8$$

$$Gain_{income} = 0.9701 - 0.8$$

$$= 0.1701$$



Student	p_i	n_i	$I(p_i,n_i)$
No	2	0	0
Yes	0	3	0

$$I(2,0) = -\frac{2}{2}\log_2\left(\frac{2}{2}\right) - \frac{0}{2}\log_2\left(\frac{0}{2}\right) = 0$$

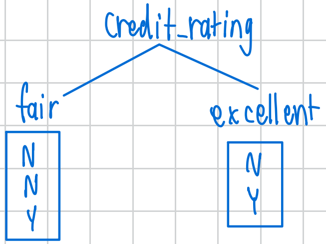
$$I(0,3) = -\frac{0}{3}\log_2\left(\frac{0}{3}\right) - \frac{3}{3}\log_2\left(\frac{3}{3}\right) = 0$$

$$Info_{Student}(D) = \frac{2}{5}I(2,0) + \frac{3}{5}I(0,3)$$

$$= \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$

$$Gain(Student) = 0.9701 - 0$$

$$= 0.9701$$



Credit	p_i	n_i	$I(p_i,n_i)$
fair	1	2	0.9183
excellent	1	1	1

$$I(1,2) = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right) = 0.9183$$

$$I(1,1) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$$

$$Info_{Credit}(D) = \frac{3}{5}I(1,2) + \frac{2}{5}I(1,1)$$

$$= \frac{3}{5}(0.9183) + \frac{2}{5}(1) = 0.9510$$

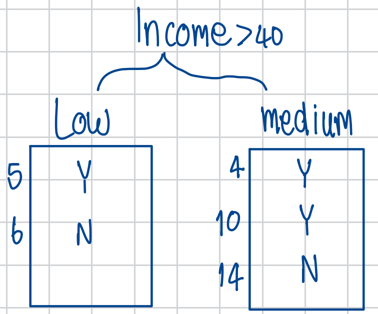
$$Gain(Credit) = 0.9701 - 0.9510 = 0.0199$$

* รู้ว่า rootnode ของ 31..40 เป็นการจัดกลุ่มตัวที่ ครบทุกตัวก็ไม่แบ่งต่อ

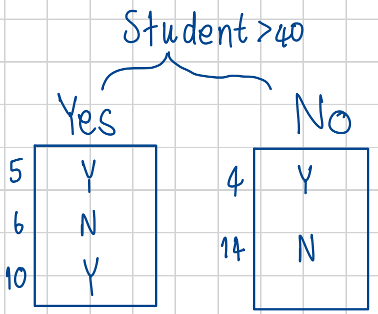
$$Info(D) = I(3,2) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right) = 0.971$$

Training data set: Who buys computer?

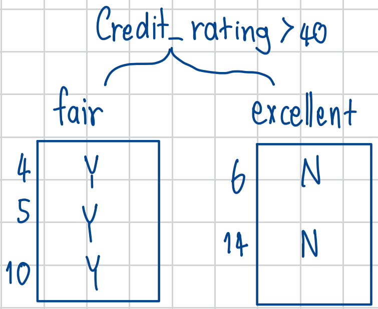
	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31…40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31…40	medium	no	excellent	yes
13	31…40	high	yes	fair	yes
14	>40	medium	no	excellent	no



ncome	p_i	n_i	$I(p_i,n_i)$
Low	1	1	1
Medium	2	1	0.918



Student	p_i	n_i	$I(p_i,n_i)$
Yes	2	1	0.918
No	1	1	1



Credit_rating	p_i	n_i	$I(p_i,n_i)$
fair	3	0	0
excellent	0	2	0

$Info_{income>40}$

$$I(1,1) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$$

$$I(2,1) = -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) = 0.918$$

$$Info_{income>40}(D) = \frac{2}{5}I(1,1) + \frac{3}{5}I(2,1)$$

แทนค่า $= \frac{2}{5}(1) + \frac{3}{5}(0.918) = 0.551$

$$Gain(Income>40) = 0.971 - 0.551 = 0.42$$

$Info_{Student>40}$

$$I(2,1) = -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) = 0.918$$

$$I(1,1) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$$

$$Info_{Student>40}(D) = \frac{3}{5}I(2,1) + \frac{2}{5}I(1,1)$$

แทนค่า $= \frac{3}{5}(0.918) + \frac{2}{5}(1) = 0.551$

$$Gain(Student>40) = 0.971 - 0.551 = 0.42$$

$Info_{Credit\_rating>40}$

$$Info_{Credit\_rating>40}(D) = \frac{3}{5}I(3,0) + \frac{2}{5}I(0,2)$$

แทนค่า $= \frac{3}{5}(0) + \frac{2}{5}(0)$

$= 0$

$$Gain(Credit\_rating>40) = 0.971 - 0 = 0.971$$

จากการหา Root Node แรก คือ พบว่า Gain ที่มีค่ามากที่สุด คือ Age เป็น 0.246 — ดีที่สุด.

ต่อมาจะหาเป็นแยกตาม Subset ใน Age แบ่งตามช่วงอายุ 3 ระดับ ผลที่ได้คือ 31..40 เป็น yes ทั้งหมด = ตอบได้เลยว่า ซื้อ/Buy
เหลือ ≥30 แยกต่อที่เหลือ income, student, credit พบว่า ≥3 Gain ของ Student มากที่สุด และคือ 0.9701 — ดีที่สุด
และหากเป็น student ตัดได้ว่า no — ไม่ซื้อ
Yes — ซื้อ

ในส่วน Age>40 พบว่า Credit_rating มีผลเป็น 0.971 — ดีที่สุด และสามารถระบุได้ 100% ในกรณี excellent = ไม่ซื้อ
fair = ซื้อ